Project 1 – Product Sales Insights

How does weather impact sales of the top three best selling products?

SQL

My SQL code was done in R and I used a relative path to make it reproducible. In my SQL code I first identified the top three bestselling products based on units sold from the train table, they are item numbers 45, 9 and 5. This query was referenced to make the pipeline more robust. I then used the key table to join sales data with the correct weather station using store_nbr. After, I wrote a query to return the daily sales and average temperature for one of the top 3 products. Lastly, I combined these steps to make a query that will be used for the R code, this query will return a data frame with only the top 3 items and all their weather variables.

R Programming

Data Cleaning

For empty strings I checked the variables that were in the character type, there were none in sunset or sunrise. Empty strings were found in codesum, I dealt with this by replacing the empty values to NA as empty strings doesn't mean no significant weather occurred, it could be that is missing from not being recorded for example. For special characters in preciptotal and snowfall, I checked if they were present and found none. Additionally, I checked the data types of these two and they are numeric, so no action was needed.

I examined the percentage of missing values for each variable. I decided to drop depart (66% missing), sunrise (58% missing) and sunset (55%) missing as imputing would create these would be inaccurate and because they did not seem as relevant to the question. I did not impute snowfall (35% missing) or codesum (55% missing) as missing data doesn't necessarily indicate no snowfall or significant weather, as it might not have been recorded. Imputing these variables would produce misleading results.

For the rest of the weather variables (having less than 5% missing data – considered safe for imputation), I used Multiple Imputation by Chained Equations (MICE) as this is considered much stronger way of imputing data as it uses other variables to predict the missing values. A seed was created so the imputation can be reproducible. Since the imputed variables had very low missing values, I used one completed dataset from MICE for practicality.

In accordance with best data science practices, I Imputed the data after doing the train-validation-test split so that later data was not used to define earlier preprocessing steps. I also imputed the data after dropping columns and after removing extreme outliers in units so that the imputation model was fitted on clean data. First, the imputation model was fitted on the training split only, and this training-fitted imputation model was then applied to the validation split. After selecting the best performing model on validation, the imputation model was refitted on the combined training and validation data, and this final imputation model was then applied to the test split. This kept the test data out of the model selection stage while allowing the final preprocessing pipeline to learn from all available pre-test data. After each imputation stage I checked the variables that were imputed for missing data, there were none.

For outliers in units, I created a simple plot to see its values (see plot document), there seemed to be two outliers. I then checked the counts for each unique value in units. Like my plot, there were two massive values for units, one at 3369 and the other at 5568. This is an absurd amount compared to the previous highest value for units at 503, this might be considered an outlier as this also seem high, but it falls within the bounds of the other values for units so it's realistic.

I did not use an IQR because units are zero-inflated, and so a generic rule base cut off could be overly aggressive and remove plausible observations. Instead, I used a domain-informed threshold of 1000. By doing so it prevents any rows where units are above 1000 from entering the DF. Effectively removing the 2 outliers.

Features

I created the features before splitting and imputation to ensure the same features would be applied to all subsets of data. The features dow (day of week), is_weekend month, and store_numbers, were from the date column and the store_nbr column. These variables had no missing values and were not part of the imputation process, so creating these features beforehand did not affect imputed values.

Dow will show what day each row is (Monday - Sunday). In retail people tend to shop more on weekends, resulting in large sale spikes over the weekends. Including dow in the model controls for this weekly pattern, reducing the chance that the weather variable is capturing the change in day-to-day demands.

Is_weekend is a binary feature and will show a 0 for weekdays and a 1 for weekends. This feature was added as it's simpler to interpret than dow. Like in dow, people tend to shop more on weekends. Including is_weekend controls for weekend spikes, reducing the chance the weather variable is capturing weekend demands.

Month will show what month each row is. In retail sales can often be seasonal, where there might be more sales in the winter for example. Including month in the model controls for this seasonal pattern, reducing the chance that the weather variable is capturing the change in month-to-month demands.

Store_number was factorised (categorised) and will show what store each row is. This is important as the model requires it to be treated as a categorical variable rather than a numeric variable. In retail, different stores can have very different baseline sales, because of a difference in size, location e.c.t. Including store_number in the model controls for these store-to-store differences, reducing the chance that the weather variable is capturing the change in store-to-store demands.

Model Validation

I used Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and R2 to evaluate model performance. RMSE measures the square root of the average squared difference between predicted and actual values, so lower values indicate better predictive accuracy. MAE measures the average absolute difference between predicted and actual values, so lower values are also better. R2 measures how much of the variation in sales is explained by the model, so higher values indicate better fit.

I used a train-validation-test split (70-10-20) to evaluate my models. Since the data is time-series data I used sequential splitting so earlier observations were used to predict later unseen observations. After the initial split, I overwrote the splits using the first validation and test dates so that the same date could not appear in more than one split. This reduces the chance of leakage across splits.

Model selection was based on the performance in the validation stage. I prioritised RMSE as the main value because larger forecasting errors are more costly for stock planning. I then used MAE and R2 as supporting checks.

As there were three items, I created separate training, validation and test splits for each item. This was done because the relationship between units sold and weather is unlikely to be the same across the three items.

I opted to use linear regression models because they're highly interpretable and fits the aim of the project. For each item I built four models. The first with units and the weather variable to get a baseline, a second model adding dow and month to control for weekly and seasonal patterns, a third model adding store number to control for store-level differences and a fourth model replacing dow with is_weekend. This was done to see if it performs as well as dow whilst being more interpretable. This modelling process was repeated for tavg and cool.

I also used decision trees as a comparison point in the validation stage. For the decision trees, I used the best performing formulas for each item and for tavg and cool from the linear models so that they could be comparable. The decision trees will also show feature importance.

The linear and tree models were trained on the imputed training split and evaluated on the imputed validation split using RMSE, MAE and R2. This allowed me to see which models performed best to unseen data. As you will see from the tables below, linear model 3 was selected as the final model because it consistently had the lowest RMSE values across the linear models, while being more interpretable and producing similar results to the tree models. Although model 4 occasionally produced a marginally lower MAE, those gains were small and came with slightly worse RMSE and R2, so model 3 provided the best overall balance of predictive power and interpretability.

Because linear regression can produce negative predicted values, I set the lowest predicted value to be zero. This is important because it makes the predictions realistic as units sold cannot be negative.

After model selection, I combined the raw training and validation data and imputed it. The test split was also imputed at this stage. The combined imputed train and validation split, and the test split was then divided for each item to allow for independent analysis. For each item, the final model used the combined training and validation split so that more data can be used to see the relationships. The test split was not used in model selection and was used in the final evaluation of model performance on unseen data.

Linear Model Validation for Tavg

Linear Validation Models for Item 45, 9 and 5 for tavg	**MAE**	**RMSE**	**R2**
lm_45_1_tavg *(units ~ tavg)*	30.852402	35.13465	-0.009435464
lm_45_2_tavg *(units ~ tavg+dow+month)*	29.916051	34.79201	0.010156932
lm_45_3_tavg *(units ~ tavg+dow+month+store_number)*	8.215297	15.95740	0.791775600
lm_45_4_tavg *(units ~ tavg+is_weekend +month +store_number)*	8.176160	16.03992	0.789616443
lm_9_1_tavg *(units ~ tavg)*	32.55408	46.21398	0.005382882
lm_9_2_tavg *(units ~ tavg+dow+month)*	32.58334	45.97863	0.015487350

lm_9_3_tavg *(units ~ tavg+dow+month+store_number)*	10.90919	22.79291	0.758059329
lm_9_4_tavg *(units ~ tavg+is_weekend +month +store_number)*	10.75017	22.87822	0.756244897
lm_5_1_tavg *(units ~ tavg)*	24.34957	32.00075	0.0006504997
lm_5_2_tavg *(units ~ tavg+dow+month)*	24.78053	31.91291	0.0061296234
lm_5_3_tavg *(units ~ tavg+dow+month+store_number)*	11.25097	19.98901	0.6100763980
lm_5_4_tavg *(units ~ tavg+is_weekend +month +store_number)*	11.20122	20.04562	0.6078648338

Linear Model Validation for Cool

Linear Validation Models for Item 45, 9 and 5 for Cool	**MAE**	**RMSE**	**R2**
lm_45_1_cool *(units ~ cool)*	30.562267	35.07421	-0.005965632
lm_45_2_cool *(units ~ cool+dow+month)*	30.171459	34.75209	0.012426933
lm_45_3_cool *(units ~ cool+dow+month+store_number)*	8.288618	16.03399	0.789771856
lm_45_4_cool *(units ~ cool+is_weekend +month +store_number)*	8.236209	16.11305	0.787693772
lm_9_1_cool *(units ~ cool)*	31.79985	46.33698	8.108713e-05
lm_9_2_cool *(units ~ cool+dow+month)*	32.26088	46.16890	7.322328e-03
lm_9_3_cool *(units ~ cool+dow+month+store_number)*	10.77599	22.82342	7.574113e-01
lm_9_4_cool *(units ~ cool+is_weekend +month +store_number)*	10.57335	22.91174	7.555301e-01
lm_5_1_cool *(units ~ cool)*	24.56542	32.02030	-0.0005705816
lm_5_2_cool *(units ~ cool+dow+month)*	25.17330	31.92179	0.0055760831
lm_5_3_cool *(units ~ cool+dow+month+store_number)*	11.23540	19.97113	0.6107735721
lm_5_4_cool *(units ~ cool+is_weekend +month +store_number)*	11.17781	20.02786	0.6085590579

Overall, across all items and for both weather variables, the best performing validation model controlled for day of week, month and store number (model 3). Compared to the models with just the weather variable or weather variable + dow + month model 3 had substantially lower MAE, lower RMSE and higher R2 values. Showing that controlling weekly patterns, seasonal

patterns, and store level differences improves predictive accuracy. The biggest improvement was adding store_number, suggesting that differences between stores explain an important proportion of variation in sales. In addition, model 4 replaced dow with is_weekend, with the aim to increase interpretability, but overall, it performed worse than model 3. I selected model 3 for further analysis as It generally had the strongest predictive power out of the linear models, and overall, lower RMSE values. Although it's slightly less interpretable, this is justified as the main aim of the report to estimate the effect of weather.

Tree Model Validation for Tavg

Tree Validation Models for Item 45, 9 and 5 for Tavg	**MAE**	**RMSE**	**R2**
tree_45_3_tavg *(units ~ tavg+dow+month+store_number)*	6.883778	15.26152	0.8095402
tree_9_3_tavg *(units ~ tavg+dow+month+store_number)*	9.617631	22.96817	0.7543243
tree_5_3_tavg *(units ~ tavg+dow+month+store_number)*	9.839639	19.64962	0.6232048

Tree Model Validation for Cool

Tree Validation Models for Item 45, 9 and 5 for Cool	**MAE**	**RMSE**	**R2**
tree_45_3_cool *(units ~ cool+dow+month+store_number)*	6.883778	15.26152	0.8095402
tree_9_3_cool *(units ~ cool+dow+month+store_number)*	9.5186	22.72815	0.7594322
tree_5_3_cool *(units ~ cool+dow+month+store_number)*	9.809937	19.48916	0.6293334

Since the best performing linear model was model 3 I used the same formula for the tree models so that they are directly comparable. Overall, the tree model performed similarly to the linear model, however its performance varied for each item. The largest difference was observed for item 45 when comparing lm_45_3_cool to tree_45_3 cool, the MAE decreased from 8.24 to 6.88, RMSE decreased from 16.11 to 15.26, and the R2 increased from 0.788 to 0.810. For item 9, the linear and tree models were most identical in performance, while for item 5, the tree model showed marginal improvement over the linear model. Although the tree models performed competitively, the linear models are much more interpretable and provide coefficients that can answer this projects' question. For that reason, linear model 3 will be used for further analysis.

Tree Model Feature importance for Tavg

Tree Models Feature Importance for Item 45, 9 and 5 for Tavg	**Store_number**	**Dow**	**Month**	**Tavg**
tree_45_3_tavg *(units ~ tavg+dow+month+store_number)*	3.793880e+07	1.181444e+06	2.160124e+02	6.241640e+04

tree_9_3_tavg *(units ~ tavg+dow+month+store_number)*	42199527.5	1431869.2	443947.3	310199.5
tree_5_3_tavg *(units ~ tavg+dow+month+store_number)*	23626931.87	815538.55	93491.17	111658.14

Tree Model Feature importance for Cool

Tree Models Feature Importance for Item 45, 9 and 5 for Cool	**Store_number**	**Dow**	**Month**	**Cool**
tree_45_3_cool *(units ~ cool+dow+month+store_number)*	3.793880e+07	1.181444e+06	2.160124e+02	3.524552e+04
tree_9_3_cool *(units ~ cool+dow+month+store_number)*	42239490.2	1431869.2	505760.1	226796.2
tree_5_3_cool *(units ~ cool+dow+month+store_number)*	23580811.63	815538.55	93491.17	54075.58

For the decision tree models, store_number had the highest feature importance by far which supports the findings from the linear model validation stage, followed by dow. The weather variables had some contribution, but their importance is much smaller. Month had the lowest importance overall but contributed more than the weather variables in some cases, particularly for item 9. Overall, this suggests that store-level differences are the strongest driver of sales, followed by day-of-week, while weather has real but small influence sales, but is not the primary driver.

Final Evaluation of Linear Tavg Models for Items 45, 9 and 5 on the Test Split

Final tavg evaluation models for items 45, 9 and 5	**MAE**	**RMSE**	**R2**
lm_45_3_tavg_final	8.532617	15.09929	0.7059037
lm_9_3_tavg_final	9.716964	19.67408	0.6942068
lm_5_3_tavg_final	9.493772	16.79475	0.5511084

Final Evaluation of Linear Cool Models for Items 45, 9 and 5 on the Test Split

Final cool evaluation models for items 45, 9 and 5	**MAE**	**RMSE**	**R2**
lm_45_3_cool_final	8.515253	15.05842	0.7074935
lm_9_3_cool_final	9.699212	19.67016	0.6943287
lm_5_3_cool_final	9.470343	16.77664	0.5520761

For both weather variables, after refitting the model on the combined training and validation data and evaluating it on the test split as a final unbiased check on unseen data, the RMSE, MAE and R2 values were similar to the validation results, suggesting that the model generalised reasonably well to unseen data.

Interpreting Key Model Outputs

Tavg Coefficients from Final Models for Items 45, 9 and 5

Final Tavg Models for Items 45, 9 and 5	Tavg Estimate	Tavg P value
lm_45_3_tavg_final	0.038419	5.57e-08
lm_9_3_tavg_final	-0.07355	8.19e-09
lm_5_3_tavg_final	0.002666	0.812996

Item 45 Tavg

Controlling for day of week, month and store number, there is strong evidence that average temperature is positively associated with units sold, with an estimated coefficient 0.038419 and a p value of 5.57e-08. This indicates that, for every 1° F increase in average temperature, expected units sold are expected to increase by about 0.038, holding all other variables constant.

Item 9 Tavg

Controlling for day of week, month and store number, there is strong evidence that average temperature is negatively associated with units sold, with an estimated coefficient -0.07355 and a p value of 8.19e-09. This indicates that, for every 1° F increase in average temperature, units sold are expected to decrease by about 0.074, holding all other variables constant.

For item 5 Tavg

Controlling for day of week, month and store number, there is little to no evidence that average temperature is positively associated with units sold, with an estimated coefficient 0.002666 and a p value of 0.812996. This effect isn't statistically significant, so it shouldn't be interpreted as meaningful evidence of a relationship.

Cool Coefficients from Final Models for Items 45, 9 and 5

Final Cool Models for Items 45, 9 and 5	Cool Estimate	Cool P value
lm_45_3_cool_final	0.042613	0.005194
lm_9_3_cool_final	0.05162	0.059648
lm_5_3_cool_final	0.103003	2.20e-05

Item 45 Cool

Controlling for day of week, month and store number, there is strong evidence that cooling degree days is positively associated with units sold, with an estimated coefficient 0.042613 and a p value of 0.005194. This indicates that, for every 1 unit increase in cooling degree days, units sold are expected to increase by about 0.043, holding all other variables constant.

Item 9 Cool

Controlling for day of week, month and store number, there is weak evidence that cooling degree days is positively associated with units sold, with an estimated coefficient 0.05162 and a p value

of 0.059648. This effect isn't statistically significant, so it shouldn't be interpreted as meaningful evidence of a relationship.

Item 5 Cool

Controlling for day of week, month and store number, there is strong evidence that cooling degree days is positively associated with units sold, with an estimated coefficient 0.103003 and a p value of 2.20e-05. This indicates that, for every 1 unit increase in cooling degree days, units sold are expected to increase by about 0.103, holding all other variables constant.

Summaries

Tavg is defined as average temperature in Fahrenheit and cool is defined as cooling degree days, which measures how many degrees a day's temperature is above 65°F. These variables help explain whether sales are influenced by average temperature or by especially hot temperature.

Item 45

For item 45, both cooling degree days and average temperature have a positive and statistically significant relationship with units sold. Tavg and cool are measured differently and from separate models so their influences are not directly comparable. Cool estimate of 0.042613 and tavg estimate of 0.038419, with p values of 0.005194 and 5.57e-08 respectively. This suggests that sales tend to increase as the weather gets warmer, especially on days above 65°F. Overall, item 45 seems to behave like a warm-weather product.

Item 9

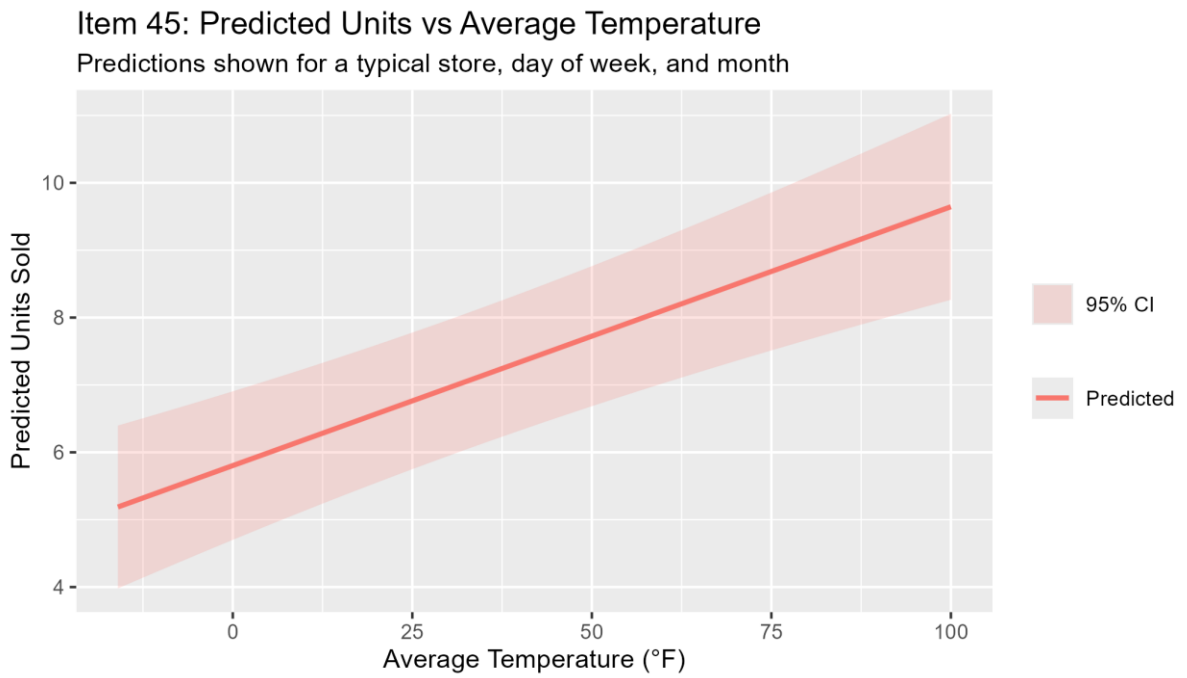
For item 9, average temperature has a statistically significant negative relationship with units sold, whilst cooling degree days have a positive but not statistically significant relationship. Tavg estimate of -0.07355 and cool estimate of 0.05162, with p values of 8.19e-09 and 0.059648 respectively. Since the cool effect is not statistically significant, it should be treated with caution. The clearest result here is that sales of item 9 tend to fall as temperatures rise, which suggests it is more of a cool-weather product.

Item 5

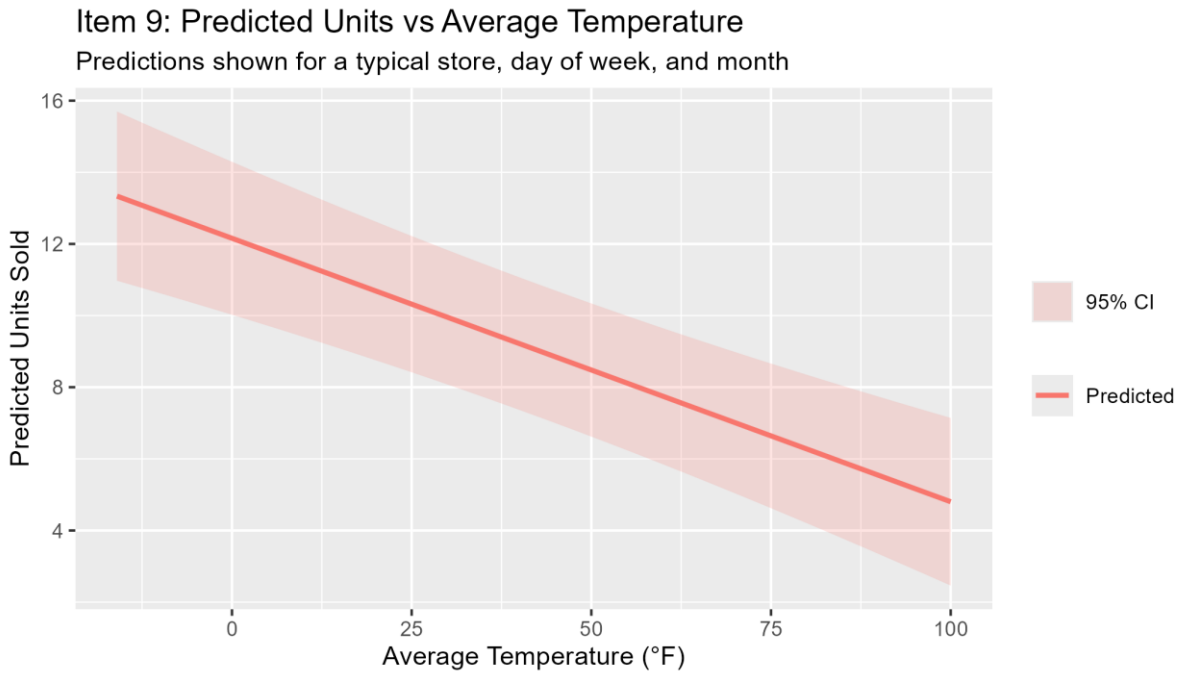
For item 5, cooling degree days has a statistically significant positive relationship with units sold, whilst average temperature does not. Cool estimate of 0.103003 and tavg estimate of 0.002666, with p values of 2.20e-05 and 0.812996 respectively. Since the tavg result is not statistically significant, it does not provide meaningful evidence of a relationship. This suggests that item 5 sells better on genuinely hot days, rather than simply on slightly warmer ones.

Visualisations

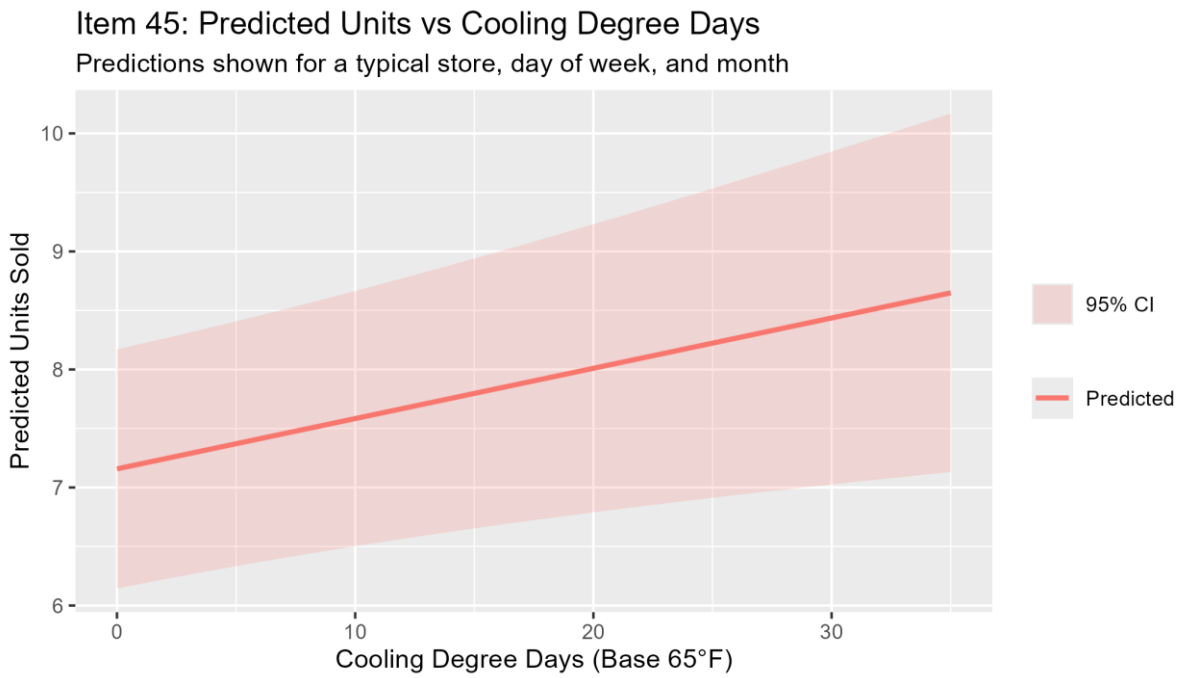
Since the estimates for item 9 cool and item 5 tavg were not statistically significant, I will not be including their plots in this report as it could be misleading, they will be included in the plot document. In addition, to make the plots more representative and less misleading I identified a typical store by first calculating the average unit's sold for every store, then selecting the store whose mean sales were closest to the median across all stores. I also chose the most common dow and month in the data to use as typical values for the other categorical predictors. These values were then used in the plotting code so the predicted sales better reflect a more representative and realistic scenario. The tree plots will be in the plot document as they were only used for the comparison in the validation stage.



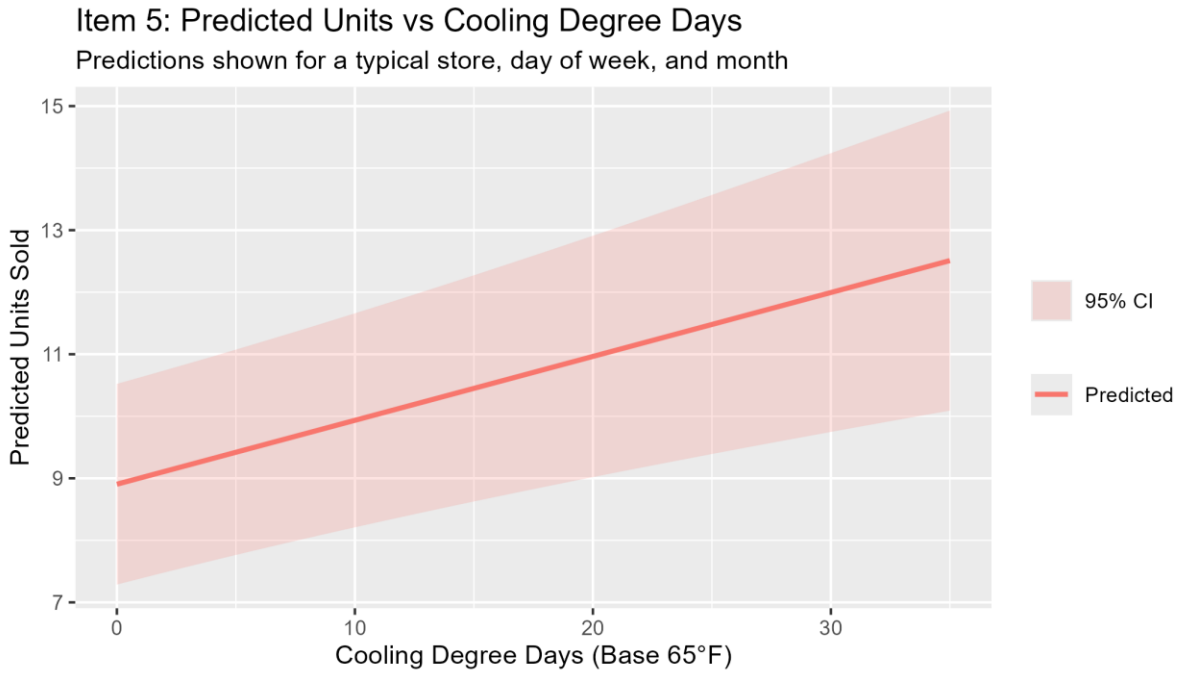
This plot shows that from a typical store, day of week and month, predicted units sold rise as average temperature increases. This supports the model result that item 45 tends to sell better in warmer weather.



This plot shows that from a typical store, day of week and month, predicted units sold falls as average temperature increases. This supports the model results that item 9 behaves like a cool-weather product.



This plot shows that from a typical store, day of week and month, predicted units sold increase as cooling degree days rise. This suggests that item 45 sells better on hotter days above 65°F.



This plot shows that from a typical store, day of week and month, predicted units sold increase as cooling degree days rise. This suggests that item 5 is especially sensitive to genuinely hot weather.

Business Recommendations

Item 45

When forecasting stocks for item 45, increase stocks when it's warmer, especially on hotter days (above 65°F.). The final models showed that both average temperature and cooling degree days have a positive and statistically significant relationship with sales. The tavg model had a coefficient of 0.038419, this means that 10°F increase in average temperature is associated with about 0.38 extra units sold per store-day. The cool model had a coefficient of 0.042613, this means that a 10-point increase in cooling degree days is associated with about 0.43 extra units sold per store-day. Across 45 stores, this is approximately 17 extra units per day for a 10°F increase in average temperature and approximately 19 extra units per day for a 10-point increase in cooling degree days, if the weather pattern is widespread. This suggests that demand rises as it becomes warmer and especially on days above 65°F. As such promotions can be used for item 45 on warmer days to reinforce this behaviour and more so for days above 65°F. This could help capture additional demand.

Item 9

When forecasting stocks for item 9, increase stocks when it's cooler and reduce stock during warmer periods. The final tavg model showed a statistically significant negative relationship between average temperature and sales, with a coefficient of –0.07355, whilst the final cooling degree model was not statistically significant. This means that a 10°F increase in average temperature is associated with about 0.74 fewer units sold per store-day. Across 45 stores, this is approximately 33 fewer units per day for a 10°F increase in average temperature if the warmer weather is widespread. As such promotions can be used for item 9 on cooler days to reinforce this behaviour, where demand is likely to be higher.

Item 5

When forecasting stocks for item 5, increase stocks when it's genuinely hotter (above 65°F) rather than just warm. The final cool model showed a statistically significant positive relationship between cooling degree days and sales, with a coefficient of 0.103003, whilst the final average temperature model had little evidence. This means a 10-point increase in cooling degree days is associated with about 1.03 extra units sold per store-day. Across 45 stores, this is approximately 46 extra units per day for a 10-point increase in cooling degree days if the hotter conditions are widespread. This suggests that demand rises mainly on hotter days above 65°F. As such, promotions can be used for item 5 on hotter days to reinforce this behaviour. And plan to increase stocks during heatwaves to reduce the risk of not having enough stock.